

AI-DRIVEN IDENTIFICATION OF MOLECULAR ENDOTYPES OF PANCREATIC CANCER-ASSOCIATED CACHEXIA USING MULTI-OMICS GRAPH ATTENTION

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ABSTRACT

Cancer cachexia is a life-threatening metabolic syndrome characterized by progressive skeletal muscle loss and is responsible for nearly 20% of cancer-related deaths. Despite its clinical importance, identifying reliable molecular biomarkers to distinguish cachectic from non-cachectic patients remains a major challenge. Existing computational approaches often treat miRNA and mRNA data separately or rely on traditional ensemble models that fail to capture underlying biological interactions.

In this work, we propose MOGA, a multi-omics graph-attention framework that integrates miRNA and mRNA expression data from the GSE75473 cohort for biomarker discovery. The model combines expression features with graph-derived regulatory information and attention-based learning to improve prediction.

Evaluated on 42 samples using nested five-fold cross-validation with SMOTE, MOGA achieves a mean AUROC of 0.755 ± 0.097 , AUPRC of 0.838 ± 0.079 , and MCC of 0.420 ± 0.357 , outperforming a Random Forest baseline. The framework identifies key miRNA biomarkers such as hsa-miR-423-3p, hsa-miR-4732-3p, and hsa-miR-320c, highlighting its potential for biologically meaningful discovery in small-sample settings. The code is available on

Index Terms— cancer cachexia, multi-omics, graph attention, miRNA, biomarker discovery

1. INTRODUCTION

Cancer cachexia (CC) is a complex wasting syndrome characterized by unintentional weight loss exceeding 5% of body mass within six months, along with progressive muscle depletion, systemic inflammation, and metabolic disruption that cannot be reversed through conventional nutritional support [1]. It affects approximately 50–80% of patients with

advanced cancer and contributes to nearly 20% of cancer-related deaths, significantly reducing both treatment efficacy and quality of life [2, 3].

Despite its severity, early detection of cachexia remains difficult due to the lack of robust molecular biomarkers and the multifactorial nature of the disease, involving tumour-derived signals, host inflammatory responses, and regulatory networks [4]. MicroRNAs (miRNAs), which regulate gene expression at the post-transcriptional level, have emerged as promising candidates due to their stability and role in muscle wasting pathways [5, ?].

The GSE75473 dataset, introduced by Narasimhan et al. [11], provides paired miRNA and mRNA expression profiles from skeletal muscle samples of pancreatic cancer patients. While the original study identified differentially expressed miRNAs, it relied primarily on statistical methods without leveraging modern machine learning techniques for integrative analysis.

Recent machine learning approaches, including Random Forests, have been applied to multi-omics data but often fail to capture biological interactions between features [8]. In contrast, graph-based models such as Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) explicitly model these interactions and have shown improved performance in biomedical tasks [9, 10]. However, many of these models are designed for large datasets and do not generalize well to small clinical cohorts.

To address this gap, we propose MOGA (Multi-Omics Graph Attention), a framework designed specifically for small-sample settings. It integrates expression data with graph topology and attention mechanisms to capture both molecular and regulatory signals, improving predictive performance and biomarker discovery.

2. METHODOLOGY

We propose a multi-stage framework integrating statistical, network, and deep learning approaches, where each stage progressively refines the representation of multi-omics data for robust classification. Expression data were obtained from

*This research study was conducted using publicly available data from NCBI GEO (GSE75473). Ethical approval was not required.

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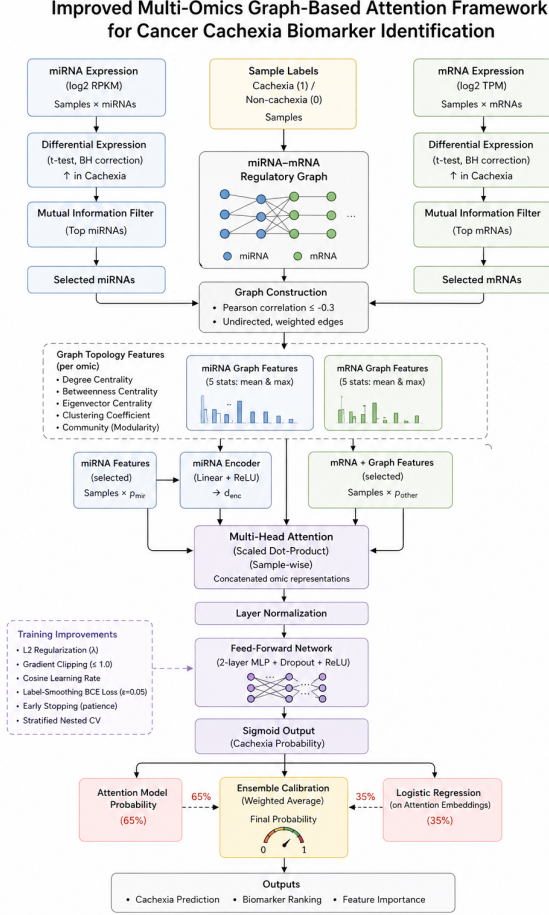


Fig. 1. Overview of the proposed multi-omics graph-based attention framework for cancer cachexia biomarker identification.

GSE75473, and corresponding labels were extracted from sample metadata. To stabilize variance and reduce skewness in the expression profiles, the data were log-transformed as:

$$X' = \log_2(X + 1) \quad (1)$$

Following preprocessing, statistically significant features were identified through differential expression analysis using Welch's t-test:

$$t = \frac{\bar{x}_1 - \bar{x}_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}} \quad (2)$$

Features satisfying:

$$\text{padj} < 0.05 \quad \text{and} \quad \log_2 FC \geq \log_2(1.4) \quad (3)$$

were retained to ensure both statistical significance and biological relevance. To further refine the selected feature set by capturing nonlinear dependencies with the target variable, mutual information was computed:

$$I(X; Y) = \sum p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (4)$$

The filtered features were then used to construct a bipartite interaction graph, enabling the modeling of regulatory relationships between omics layers. Pearson correlation was employed to quantify pairwise associations:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y} \quad (5)$$

To emphasize biologically meaningful inhibitory interactions, only negatively correlated edges ($r \leq -0.3$) were retained, resulting in a sparse and interpretable graph structure. The constructed graph and selected features were subsequently fed into a dual-encoder architecture, where modality-specific encoders learn latent representations:

$$h_m = \text{ReLU}(X_m W_m + b_m), \quad h_o = \text{ReLU}(X_o W_o + b_o) \quad (6)$$

These representations were integrated using an attention mechanism to capture cross-modal interactions:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

The aggregated representation was then passed through fully connected layers to obtain the final prediction:

$$\hat{y} = \sigma(W_2 \cdot \text{ReLU}(W_1 h + b_1) + b_2) \quad (8)$$

The model was trained using binary cross-entropy loss:

$$\mathcal{L} = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})] \quad (9)$$

To improve convergence and avoid local minima, a cosine annealing learning rate schedule was employed:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos(\pi t/T)) \quad (10)$$

Finally, model performance was assessed using a nested cross-validation strategy (5 outer folds and 3 inner folds) to ensure unbiased evaluation, while class imbalance was handled using SMOTE within the training folds.

Table 1. Performance comparison between proposed model and Random Forest

Metric	Proposed Model	Random Forest
AUROC	0.755 ± 0.097	0.726 ± 0.239
AUPRC	0.838 ± 0.079	0.753 ± 0.209
F1-score	0.667 ± 0.220	0.654 ± 0.138
MCC	0.420 ± 0.357	0.272 ± 0.275
Sensitivity	0.680	0.670
Specificity	0.700	0.600

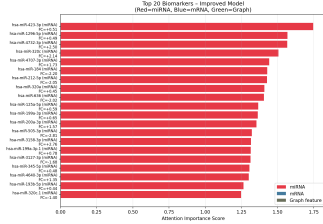


Fig. 2. Proposed Model

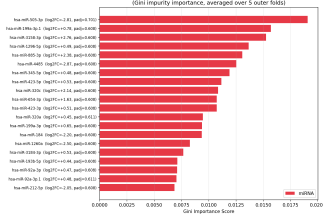


Fig. 3. Random Forest Model

3. RESULTS

The proposed framework achieved improved performance on the GSE75473 dataset. MOGA obtained a mean AUROC of 0.755 ± 0.097 and AUPRC of 0.838 ± 0.079 . It also showed better balance in classification, with an F1-score of 0.667 ± 0.220 and MCC of 0.420 ± 0.357 . Compared to the Random Forest baseline, which showed higher variance and lower robustness, the proposed model demonstrated more stable and consistent performance across folds. Comparison table is shown in Table 1.

The model identified biologically relevant miRNAs including hsa-miR-423-3p, hsa-miR-4732-3p, and hsa-miR-320c, validated from [11] suggesting meaningful biological insights beyond standard statistical methods. Top 20 biomarkers found out from both the models are shown in Fig. 1. and Fig. 2.

4. CONCLUSION

We presented MOGA, a multi-omics graph-attention framework for cancer cachexia biomarker discovery. By integrating expression data with regulatory network information, the model achieved improved predictive performance compared to traditional methods.

Importantly, the framework identified biologically relevant miRNAs that were not strongly highlighted by conventional analysis, demonstrating its ability to uncover deeper regulatory patterns.

While results are promising, the study is limited by the small sample size. Future work will focus on validation across larger datasets and incorporating additional omics layers to further improve robustness and interpretability.

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